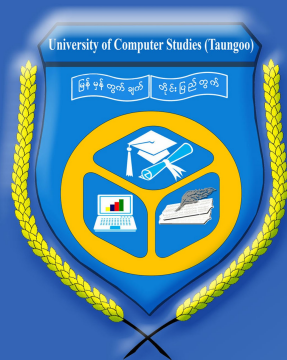


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Education Science

Local University-Industry Linkage and Job Perception

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ABSTRACT: Many universities organize internship and industrial placement for the students as part of the undergraduate learning. Internship is a temporary work experience wherein a student performs pre professional work in support of an organization's goals. Internships also allow students to connect their practical experiences in the workplace with the theoretical constructs that they have explored in the classroom. This study presents the results relating to Computer University's use of internships in their assurance of learning efforts and describes the successful attempt to use internships as the key platform for its well-developed assurance of learning program.

Keywords: University-industry linkages, Internships, Industrial placement

1. INTRODUCTION

The information and communication technology (ICT) profession includes a very wide range of job titles, designations and descriptions, with a vast array of university programs existing to supply graduates for the underlying roles. In addition to input into the curriculum, the ICT industry may engage more directly with learning and teaching via activities such as supporting forms of work integrated learning (WIL) including internships and industry-related projects, and there is evidence to support the value of these interactions [6] [7]

The ICT industry has long been characterized by skill shortages [1] [11] [9], with relatively low numbers of students enrolling in ICT degrees seen as a major contributing factor [5] [2]. Poor perceptions of university ICT courses and ICT jobs have contributed to the enrolment issue [17] [13] [3] and it has been argued that better relationships between industry and university are needed [8] [3] [10] to help address this problem.

The ICT industry in its various forms provides the destination for ICT graduates. Universities therefore rely on the ICT industry to communicate its need in terms of student knowledge and skills, and to support student attainment of these by providing opportunities for students to engage with industry whilst still studying [16].

Stronger U-I linkages have been called for by both the ICT industry and by universities. Industry would like further involvement in Curriculum design, both directly and through professional associations. The stronger relations should also help improve perceptions of ICT as a career, ensuring that students have more accurate perceptions of the careers for which they are being prepared and

ultimately helping to reduce skill shortages by improving the student experience and therefore reducing attrition.

The traditional universities cannot fulfill the requirements as demanded by modern industrial manufacturing processes and businesses. So, University Industry linkages (U-I) stimulate new scientific and technological research activities when mutual bonding between the two becomes stronger.

From industry perspective, U-I is very fruitful in terms of technological development, lower cost of research, cheap labor, and pool of expertise at one place. From university perspective, academicians get to know the latest demand of the local, national and international market, according to which, they grab and improve the knowledge and transfer it to the students through improved pedagogy [15]. Thus U-I creates a win-win situation between universities and industries [12].

University and industry have different intentions and purposes but somehow, they are related to each other. Universities are more student and faculty focused through research and technological innovation while industries are more concerned to achieve their sales and revenues. But due to competition, industries need skilled labor and technological innovation in their products while universities want to research in the areas of market demand.

Universities have emerged as central actors in knowledge based economies and are expected to play pivotal role in sprucing technological change and innovation. The concept of 'teaching university' has far gone in developed countries, and they have augmented towards 'entrepreneurial university' [4].

1.1 Internship Program

In University of Computer Studies (Taungoo), since 2016- 2017 academic year final year students must take internship program. It takes for sixteen weeks continuously, integrated project, mini thesis become compulsory for every ICT students before their graduation. The student will be added to the list of applicants interested in the internship field. The internship coordinator sends the internship request letters to the ICT companies. Upon request, the internship coordinator will assist prospective employers in selecting and accepting students.

Interns are integrated into a team working: working with team from different universities or working together with experts in the respective field. It provides the impetus for the students to comprehend and appreciate real-life working experiences. Students may realize their ambition and ascertain their career path from the experience gained during industrial training. The objectives of the internship program are as follows:

- To provide an opportunity for students to discover, learn about, and familiarize with industry of their discipline, and with organizations within the industry. Students will acquire interpersonal skills through meeting with professionals in their field of study.
- To provide an opportunity for students to observe real-life practices and implementation of theoretical lessons and principles. Students will acquire practical skills and experience working on projects and alongside industry experts.
- Overall, students will get experience as following:
 - ❖ Organizational skills and professional awareness.
 - ❖ Ability to work under supervisor and directions.
 - ❖ Efficiently completing tasks, fostering good relationship.
 - ❖ Communication skills and contribution to other people.[10]

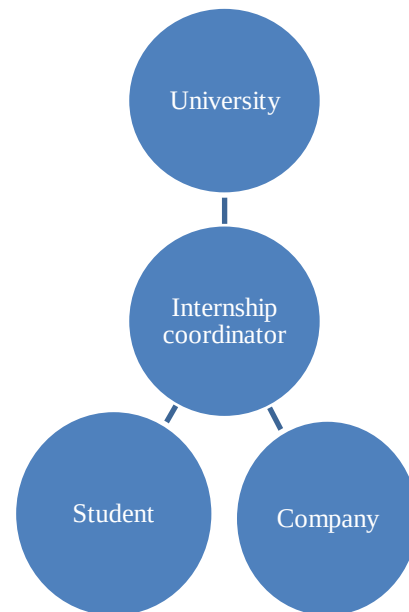


Figure 1. Role of Internship Coordinator

1.2 Student Intern's Responsibilities

To be available, and on time, for schedule activities.

- ✓ To notify both company supervisor and coordinator in advance of necessary advance.
- ✓ To plan and carry out all assignments in professional manner.
- ✓ To record logbook of daily activities throughout the internship.
- ✓ To submit the final report by the deadline specified by the internship coordinator.
- ✓ To attend internship seminar as scheduled and to maintain regular communication with the internship coordinator.

1.3 Company's Supervisor Responsibilities

- ✚ To orient the student to the working environment and to establish a schedule of work activities.
- ✚ To provide the intern with meaningful work experience through the internship.
- ✚ To conduct with the student periodic evaluation sessions to assess the student's performance and progress, to review the

projected work schedule, and to discuss matters of the students.

- ✚ To notify the coordinator immediately of any problems arising out of association with, or supervision of the student.
- ✚ To submit a final written evaluation of intern's performance.

Evaluation is an essential part of an internship. An evaluation helps the intern to acknowledge work strengths and areas for improvement.

- Internship Provider's Evaluation (by employer handbook)
 - Attitude (20%)
 - Skill (40%)
 - Communication (15%)
 - Overall Performance (15%)
- Attendance (10%) (by Logbook)
- Internship Report
- Oral Presentation

1.4 Internship Documentation Logbook

Logbook contains a record or proof of a student's learning accomplishments, learning experience, and learning outcomes. Students can include their logbooks in their Curriculum Vitae for future, job seeking purposes. Students must provide details of their internship including:

- Day and time of activities;
- Activities and duties performed;
- Reactions and comments from their supervisor/industrial tutor.

1.5 Internship Documentation Handbook

Employer handbook contains the information of responsibilities for student and company, certificate of learner internship, progress report of intern and survey of internship.

1.6 Internship Report

There are various types of report: practical, field-work, laboratory, technical etc. Generally, report is a systematic and well-organized document that defines and analyzes a subject or a problem. For example, engineering students are often

required to write a progress or technical report as part of their internship/placement assessment students to submit an internship report as one of the assessment tasks for their internship programs.

Generally, students are required to write eight to nine pages that covers:

- The overview of the company;
- Description of the student's position(s) and role(s) – technical and non-technical work, training received and skills learned, special assignment(s), and etc;
- Major projects and distinctiveness of the internship/placement;
- Applicability of internship/placement to the student's degree and program;
- Effects of internship/placement on the student's employment/career goals;
- Students' overall thoughts of the internship/placement and the company.

1.7 Oral Presentation

In oral presentation, students demonstrate not only their understanding and knowledge of their internship/placement outcomes, but also their ability to effectively communicate their ideas to a group of audience. It is also common that the student's presentation is followed by a discussion section, where the supervisor may ask questions and the student needs to respond and defend any presented materials.

The presentation should include some pictures of the work done in internship/placement, i.e., the job site, the co-workers, product samples, and etc. Students are reminded to ask for the company's permission for using these pictures.

1.8 Issue of Internship

- Sometimes the company and the intern, or the experience and the intern are not a good fit.
- Students and supervisors may contact the internship coordinator at any time during the internship to discuss any issues that may arise.

The use of internships is a powerful learning tool that allows business students to make connections between their classroom experience and the world of work. If designed appropriately and positioned correctly in the curriculum, they can also be an ideal opportunity to conduct the learning activities with industry needs.

The reasons for universities to seek cooperation with industry are: i) funds, ii) exposure, iii) real world problems, vi) training, v) industry visit, vi) internship, vii) problem based learning, viii) on job training, xi) scholarship program, x) job opportunity, xi) current market needs and xii) joint research.

For industry, the reasons to seek cooperation with universities are:

- Access to manpower, well-trained graduates knowledgeable faculty;
- Access to basic and applied research results;
- Solutions to specific problems or professional expertise;
- Access to university facilities, not available in industry;
- Assistance in continuing education and training;
- Obtaining prestige or enhancing the company's image
- Being good local citizens or good community relations.

2. DATA

This paper expresses a case study on University of Computer Studies (Taungoo) with 29 ICT industries linkages. In order to examine industry perceptions of involvement in ICT with university, the opinions of a range of employers in the ICT industry were sought via a survey. Data for the research is collected by the researcher from industry side by questionnaires of employer's handbook was shown in Table 1.

The study consists of 122 internship students, final year computer science students and computer technology students from University of Computer Studies (Taungoo). This questionnaire was sent to 29 IT industries by handbook.

The data of student's enrolment and graduation in 2016- 2017 and 2017- 2018 academic years is collected from Department of Student Affairs (University of Computer Studies (Taungoo)). Then the employment of graduates in ICT industries is conducted by the survey of

graduate students of these two academic years. The data of graduate students, attending further studies, not getting jobs yet and employing in non ICT industry is also collected. By using these data, the graduate employment rate of two academic years was calculated was shown in Table 2.

3. RESULT AND DISCUSSION

A total of 29 ICT industries to the survey upon 122 intern students were received: 57.4%, resourcefulness percentage of students was observed as the least among 11 characteristics of employer's evaluation. And technical skill was the second least characteristics found in only students of 68.9%. It focus the necessary of students more practical skills and work experiences. The university students should be sent to industry when they are 2nd year, 3rd year and 4th year students in order to familiar with real-life practices. So universities require to improve their connections with industry by mechanisms such as using industry advisory boards for curriculum development, and by offering short courses to industry to provide a good two-way communication channel between academia and industry.

Large percent of students (71.3%) was characterized in the cases of initiative and professionalism respectively. The students of 83.6% were recommended by intern providers as positive thinking and completing tasks students. Providers agreed that 84.4% were good at team works and 78.7% have performed overall works well. According to above data, most of intern students from universities have positive behaviors, good time management, diligence, adaptability and fine collaboration in terms of work readiness skills.

The ICT industry has long been characterized by skill shortages with relatively low numbers of students enrolling in ICT degrees seen as major contribution factor. The large numbers of students, 70.9% and 84.6% were enrolled in ICT degrees in academic year 2016-2017 and 2018-2019 respectively. It provide the fact that an ICT university has produced large number of ICT professionals yearly, so ICT industries can overcome labor shortage problem but they need to provide more intern spaces and guaranteed work placements to students. Poor perceptions of university ICT courses and ICT jobs have contributed to the enrolment issue and it has been argued that better relationships between industry and universities are needed to help address this

problem. Therefore the relationship between university and industry engagement are needed to tie stronger.

And enrolled students of 74.5% to 67.6% have got graduation, and 33.3% and 34.0% of students who undertake internship program as a part of their degree were employed in ICT industry. Paper confirms University students who undertake University Industry Linkages as part of their degree consistently achieve better employment outcomes.

While the value of industry involvement with ICT in universities is widely recognized by ICT academics, graduates and industry, this study identified issues that could potentially be improved by stronger U-I linkages, and called for them to be addressed. These issues include shortages of ICT professionals, poor perceptions of ICT as a course of study and as a career, and gaps in the knowledge and skills of graduates.

Table 1. Employer's Evaluation

Characteristics	Excellent	Above Average	Average	Less than Average	Total	Agreement (%)
Initiative	27	60	30	5	122	71.3
Job judgment/ Maturity/ Professionalism	16	71	33	2	122	71.3
Resourcefulness	13	57	42	10	122	57.4
Positive thinking	45	57	16	4	122	83.6
Technical skill	30	54	31	7	122	68.9
Completing tasks	41	61	15	5	122	83.6
Ability to learn new concept and ideas	38	54	23	7	122	75.4
Can work independently	14	80	21	7	122	77.0
Team work	54	49	16	3	122	84.4
Communication skill	47	48	19	8	122	77.9
Overall performance	33	63	23	3	122	78.7

Note: Responses Range from Excellent to Less than Average

Table 2. Graduate Employment Rate in ICT Industry

Academic Year	Enrolment (%)	Graduate (%)	Employment (%)	Other (%)
2016-2017	70.9	74.5	33.3	66.7
2017-2018	84.6	67.6	34.0	66.0

4. CONCLUSION

This study evaluates the current U-I situation in computer university, by elaborating the perspective of both university and industry sides.

Although this study is very important, it still it has some limitations. Research data comes from the very limited number of industries and which is one of the limitations of this study. There are about 350 ICT company members in Myanmar Computer Industry. A case study on University of Computer Studies (Taungoo) expresses only 29 ICT industries support internship program to this university students because most universities are far from the cities that the ICT industries saturated. Most ICT industries are established in main economic cities, Yangon, Mandalay and Naypyitaw. So there are still many difficulties for students from ICT Universities to reach proper internship program.

To make sense of this study results, researchers need to expand the research data for future. The results confirm how important relationships with universities are seen to be and that industry would like to expand its level of involvement. Industry would like further involvement in curriculum design.

University could engage with industries in between academic's research and projects, internship program, industrial visit, on job training, scholarship program, training program and etc. And more ICT industries collaboration with universities,

for example in the case of internship program is required for more job opportunities of graduates. In order to measure the current situation of UI linkages in Myanmar, author will collect data regarding with UI linkages from universities as future work.

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Impact of Teaching Method on Academic Performance: An Experimental Research Study

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ABSTRACT: *This paper studied the effect of teaching method on academic performance in teaching and learning of University of Computer Studies (Taungoo). The study analyzed the teaching styles of each subject and the student assessment data of 400 students attending from first year to fifth year students between 2017-2018 and 2018-2019 academic years. The research data were evaluated and monitored by using two indicators: teaching method indicators and academic performance indicators. Qualitative data was thematically coded and quantized then entered in statistical package for social science (SPSS) alongside the quantitative data then was analyzed descriptively and inferentially and presented using statistical tables. Result of the research indicates that teaching method has an impact to academic performance.*

Keywords: Teaching method, Performance, Impact, SPSS

1. INTRODUCTION

A teaching method comprises the principles and methods used by teachers to enable student learning. These strategies are determined partly on subject matter to be taught and partly by the nature of the learner. The approaches for teaching can be broadly classified into teacher centered and student centered.

Both of the teaching methods are used in University of Computer Studies (Taungoo) based on the teachers, subjects and learners. Some teachers use hybrid method. And then, the most teaching pedagogy is Project Based Learning (PBL).

1.1 Research Motivation

The author has been teaching two computer science subjects in one academic year. But the performance of the students of these two subjects is different. In one subject, the performance of the students does not improve year after year. In another subject, the performance of the students improves year after year. Therefore this research analyzes the significant differences between the effectiveness of different teaching methods on students' academic performance.

1.2 Research Objective

The aim of this study is to enhance the student's academic performance in every subject of UCS (Taungoo).

It analyzes six subjects taught by seven teachers in Application Department. The subjects are C#, JavaScript, Computer Graphics (CG), Information Security (IS), Hyper Text Markup Language (HTML), Marketing.

Firstly, finding of the current study shows that even the computer science subjects has an impact based on teaching method and methodology.

In the context of education, good

explanation in teaching is essential for unlocking the students' understanding of the subject. From a learning perspective, explanation holds a special place as one of the core critical thinking skills (Facione 1990). Secondly, the current study examines that even good explanation has an impact based on teaching method and methodology.

Finally, the study indicates that student centered pedagogy such as Project based learning has a better effectiveness on academic performance.

2. LITERATURE REVIEW

In Teacher-Centered Approach to Learning, Teachers are the main authority figure in this model. Students are viewed as "empty vessels" whose primary role is to passively receive information (via lectures and direct instruction) with an end goal of testing and assessment. It is the primary role of teachers to pass knowledge and information onto their students. In this model, teaching and assessment are viewed as two separate entities. Student learning is measured through objectively scored tests and assessments [4].

In Student-Centered Approach to Learning, while teachers are the authority figure in this model, teachers and students play an equally active role in the learning process. The teacher's primary role is to coach and facilitate student learning and overall comprehension of material [6]. Student learning is measured through both formal and informal forms of assessment, including group projects, student portfolios, and class participation. Teaching and assessments are connected; student learning is continuously measured during teacher instruction. Commonly used teaching methods may include class participation, demonstration, recitation, memorization, or combinations of these [4].

Teacher-Student Interactive Method applies the strategies used by both teacher-centered and student-centered approaches. The subject information produced

by the learners is remembered better than the same information presented to the learners by the lecturer (Jacoby, 1978; McDaniel, Friedman & Bourne, 1978; and Slamecka & Graf, 1978). The method encourages the students to search for relevant knowledge rather than the lecturer monopolizing the transmission of information to the learners. As such, research evidence on teaching approaches maintains that this teaching method is effective in improving students' academic performance (Damodharan & Rengarajan, 1999).

Project Based Learning (PBL) is a teaching pedagogy in which students learn by actively engaging in real-world and personally meaningful projects [3]. Students work on a project over an extended period of time – from a week up to a semester – that engages them in solving a real-world problem or answering a complex question. They demonstrate their knowledge and skills by creating a public product or presentation for a real audience [1].

SPSS is short for Statistical Package for the Social Sciences, and it's used by various kinds of researchers for complex statistical data analysis. SPSS is used by market researchers, health researchers, survey companies, government entities, education researchers, marketing organizations, data miners, and many more for the processing and analyzing of survey data [2].

There are a handful of statistical methods that can be leveraged in SPSS, including:

- Descriptive statistics, including methodologies such as frequencies, cross tabulation, and descriptive ratio statistics.
- Bivariate statistics, including methodologies such as analysis of variance (ANOVA), means, correlation, and nonparametric tests.
- Numeral outcome prediction such as linear regression.
- Prediction for identifying groups, including methodologies such as cluster analysis and factor analysis.

3. RELATED WORK

In "Effect of Project-Based Learning on Students' Attitude Towards English Classes", by Burcu Duman, the study group of the research consisted of 51 students attending grade 10 at an Anatolian high school in the Western Black Sea region of Turkey. This research was analyzed by SPSS analysis tool and it showed that PBL generally improves the attitude of the students towards English classes [1].

"The Effect of Project-Based Learning on Student Performance: An Action Research Study" was studied by Darren Hiwamoto in psychology class. In this study, data analyzed found that the experimental group that engaged the project and took responsibility for the learning of their peers scored significantly higher on the multiple-choice exam when compared to the control group by using key performance indicators such as high

self-efficacy, high level of perceived control, and growth mindset [3].

"A Comparison of Teacher-Centered and Student-Centered Approaches in Educational Settings" by Hamdi Serin analyzed that Teacher has the ultimate authority and is in charge of learning for that reason students do not have adequate opportunities to develop their critical thinking and problem solving skills [4].

"Relationship Between Teaching Method And Students' Performance In mathematics In Public Secondary Schools In Dadaab Sub County, Garissa county; Kenya", was studied on all head teachers, all mathematics teachers and two students by Joel Silla Muema. And then, a total of 155 respondents were selected. In this research, SPSS analysis tool was used and teaching using ICT's was strongly correlated to students' achievement than traditional methods of teaching in mathematics [5].

"The Relationship Between Student Engagement and Their Academic Achievement" by Selim comprised of 3,268 15-year-old students from 121 U.S. schools. Multilevel analysis was run on SAS 9.2. This study suggested that educators, policy makers, and the research community need to pay more attention to student engagement and ways to enhance it [7].

4. RESEARCH DESIGN AND METHODOLOGY

In University of Computer Studies (Taungoo), Computer science and technology subjects are major subjects and others subjects such as Myanmar, Physics, Computing Mathematic and English are minor subjects.

The subjects: C#, JavaScript, Computer Graphics (CG), Information Security (IS), Hyper Text Markup Language (HTML) and Marketing were coded as C#, JS, CG, IS, HTML and M.

According to teaching styles and student's academic performance in UCS (Taungoo), the study collected the data on teaching styles in seven subjects as teaching method indicators and then corresponding assessments on 400 students from first year to fifth year within two academic years as academic performance indicators.

Five teaching method indicators are used as following in each subject.

- T1. The percentage of explanation by teachers
- T2. The percentage of self-study by students
- T3. The percentage of practice by students
- T4. The percentage of student involvement in teaching
- T5. The percentage of group work and project by students

Using the above questionnaire collects the data for the study of teaching method indicator from each teacher as shown in table 4.1.

Student performance assessment according to academic performance in UCS (Taungoo) is indicated as following.

- P1. Assessment on self-study skills
P2. Assessment on practical skills
P3. Assessment on communication skills
P4. Assessment on critical thinking skills
P5. Assessment on collaboration skills
P6. Assessment on real world problem solving skills

The performance data for the study were generated from students' academic performance assessment test scores. These scores were thematically categorized, coded and quantized by four variables: "Bad", "Natural", "Good" and "Better" as "0", "1", "2" and "3" respectively as shown in table 4.2.

Table 4.1. The percentage of teaching method indicator on each subject

Teaching Style	C #	JS	CG	IS	HT ML	M	SAD
T1	25	30	55	20	30	40	30
T2	10	10	5	20	10	20	10
T3	25	25	30	20	20	0	10
T4	15	20	10	20	20	20	20
T5	25	15	0	20	20	20	30

Table 4.2. Encoded performance assessment on each subject

Performance Indicator	C#	JS	CG	IS	HT ML	M
P1	1	1	0	2	1	1
P2	3	3	1	2	2	0
P3	2	2	1	3	2	2
P4	3	3	1	2	2	2
P5	3	3	0	3	2	2
P6	3	3	0	3	3	3
Total	15	15	3	15	12	10
Total Percent-age	83.3	83.3	16.7	83.3	66.7	55.6

5. RESEARCH FINDINGS

The study sought to establish the methodologies adopted by teachers in different teaching styles. Descriptive analysis was done to obtain minimum and maximum scores in teaching method indicator (T_i), the means of each T_i and the standard deviations. Table 4.1 summarizes the descriptive analysis of each T_i for seven subjects.

The results in tabular form of table 5.1 by SPSS analysis, the maximum value in Computer Graphics (CG) and Marketing (M), "55" and "40" are the percentage of explanation by teachers. And the standard deviation of these two subjects is the most one. The minimum value in the percentage of group work project in CG and the percentage of practical work in Marketing (M) subjects is the main key point that makes the most standard deviation.

Table 5.1. Descriptive Analysis of Teaching Method Indicator on Each subject

Subjects	N	Minimum	Maximum	Mean	Std. Deviation
C#	5	10	25	20	7.1
JS	5	10	30	20	7.9
CG	5	0	55	20	22.6
IS	5	20	20	20	0.0
HTML	5	10	30	20	7.1
M	5	0	40	20	14.1

Table 5.2. Pearson correlation analysis of teaching method on student's academic performance results (PR)

		T1	T2	T3	T4	T5	PR
T1	Pearson Correlation	1	-.457	.038	-.668	-.836	-.872
	Sig. (2-tailed)		.363	.943	.147	.038	.023
	N	6	6	6	6	6	6
T2	Pearson Correlation	-.457	1	-.778	.683	.560	.440
	Sig. (2-tailed)	.363		.068	.135	.248	.383
	N	6	6	6	6	6	6
T3	Pearson Correlation	.038	-.778	1	-.570	-.436	-.257
	Sig. (2-tailed)	.943	.068		.238	.388	.623
	N	6	6	6	6	6	6
T4	Pearson Correlation	-.668	.683	-.570	1	.683	.751

	Sig. (2-tailed)	.147	.135	.238		.135	.085
	N	6	6	6	6	6	6
T5	Pearson Correlation	-.836*	.560	-.436	.683	1	.888*
	Sig. (2-tailed)	.038	.248	.388	.135		.018
	N	6	6	6	6	6	6
PR	Pearson Correlation	-.872*	.440	-.257	.751	.888*	1
	Sig. (2-tailed)	.023	.383	.623	.085	.018	
	N	6	6	6	6	6	6

The above tabular form of table 5.2 shows Pearson's correlation. The Pearson's correlation coefficient (r) is a measure of the strength of the association between the two variables. Firstly, Students' performance results has a positive impact of the teaching method indicator T2 (self-study) in $r = 0.440$, T4 (student involvement) in $r = 0.751$ and T5 (group-work project) in $r = 0.888$. These are the factors of student-centered approach. Among them, Secondly, T5, the Project Based Learning (PBL) pedagogy has the better influence on students' performance, $r = 0.888$. And then thirdly, $r = -0.872$ shows that good explanation has a negative impact on students' performance.

Finally, the involvement in teaching and learning percentage shows that only the involvement person improves in academic performance. The percentage of teacher involvement in Computer Graphics subjects is more than student involvement. Therefore, the performance of teacher improves year after year than that of students. In PBL based subjects the performance of student's improves year after year. This finding shows that involvement in teaching and learning has a direct relationship of the academic performance.

6. CONCLUSION

The research data were evaluated and monitored by using two indicators: teaching style indicators and academic performance indicators and analyzed by SPSS analysis tools. The study established that Project Based Learning could cause an improvement on academic performance. In the computer science subjects without projects, the students emphasize only on the exam marks and they are still lack of use the respective skills in real world problems. Moreover, the study indicates that the performance of students in

projects improves year after year. It is also evidence that some computer science subjects need to use "self-study by students", "student involvement" and "PBL approach rather than to get good results on exam marks.

7. FURTHER EXTENSIONS

According to these findings, Project Based Learning approach in Computer Graphics subject will be used in next years. And then, the comparison of this subject based on no PBL and PBL will be examined as further extensions.

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The Role of Capacity Assessment in Research Capacity Building Strategy: An Experimental Study

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ABSTRACT: *This paper focuses on the role of capacity assessment in initiating research capacity building for developing institutions. Now, all the universities emphasize on University Research (UR). In University of Computer Studies (Taungoo), each faculty also initiates UR. This study is valuable to faculty member who are new to UR. It is an experimental study of one faculty of UCS(Taungoo). The experimental action results are evaluated and monitored by two indicators: capacity building process indicators and research performance indicators. It shows that capacity assessment and capacity building on target research area is a novel and alternative approach to get better performance in UR.*

Keywords: Research Capacity Building, Capacity assessment, University Research, Strategy

1. INTRODUCTION

All the universities are under pressure by UR. Some of the universities are already engagement in UR and many research professionals are there. But some are initiating UR and human resource for research management is still required. Moreover, strategy for research capacity building is vital for those initiating UR Universities. In addition, developing nations are adopting strategies in the form of capacity building, to avoid becoming perpetually dependent on international aid. The study analyzed the UR developed by seven teachers in Application Department.

Research management in the form of UR is being initiated in UCS(Taungoo) since 2017. Research Management (RM) is a part of research process. Although UCS(Taungoo) has been established some RM process, there was weakness in strategy of RCB.

1.1 Research Motivation

Research Capacity Building is ongoing process in UCS(Taungoo). It is needed to balance between the capacity building process and the results of research performance. Earlier duration of the capacity building process had not occur these balancing. Therefore the study examined how to make balancing the research capacity building results.

1.2. Research Objective

The aim of this study is to enhance Research Capacity Building (RCB) for all professionals who are new to UR and motivate them.

2. LITERATURE REVIEW

Research Management [1] involves the capabilities that research-focused organizations harness so as to optimize and amplify the processes and impacts of their research activity and outcomes. Research

Management consists of, inter alia: research planning, strategy and policy development; research ethics and integrity researcher development; partnerships and collaboration; research funding; managing funded research; research data and research information management; research uptake and utilization; organization and delivery of a research management service.

Capacity building (CB) or capacity development [6] is the process by which individuals and organizations obtain, improve, and retain the skills, knowledge, tools, equipment, and other resources needed to do their jobs competently. Capacity Building within the context Research is defined as enhancing the abilities of individuals, organizations and systems to undertake and disseminate high quality research efficiently and effectively [4] [5]. There are 3 levels of Capacity Building [3]:

- Individual: involving the development of researchers and teams via training and scholarships, to design and undertake research, write up and publish research findings, influence policy makers, etc.
- Organizational: developing the capacity of research departments in universities, think tanks and so on, to fund, manage and sustain themselves.
- Institutional: changing, over time, the 'rules of the game' and addressing the incentive structures, the political and the regulatory context and the resource base in which research is undertaken and used by policy makers.

Research Capacity Building process includes four steps [3]:

1. Capacity assessment: this step is primarily concerned with identifying the main strengths and weaknesses of the research and institutional framework at the individual, organizational and institutional levels.
2. Strategies and plan: this step involves planning the detailed activities required to deliver the programmed outcomes; costs and timescales and monitoring and evaluation arrangements which will include organizational mapping and

establishing a capacity baseline. Steps one and two would normally be undertaken and reported on during the inception stage of a research programmed.

3. Implementation: this section sets out the key roles of the research partners in supporting processes and highlights some examples of actions at each of the three levels-individual, organizational and institutional-which can contribute to effective CB.
4. Monitoring and evaluation: this section sets out the key principles to be followed in monitoring and evaluation, as well as some examples of indicators which may be used to judge the effectiveness of CB.

A performance indicator or key performance indicator (KPI) is a type of performance measurement. KPIs evaluate the success of an organization or of a particular activity (such as projects, programs, products and other initiatives) in which it engages. Since there is a need to understand well what is important, various techniques to assess the present state of the business, and its key activities, are associated with the selection of performance indicators. These assessments often lead to the identification of potential improvements, so performance indicators are routinely associated with 'performance improvement' initiatives [2].

3. RELATED WORK

Jayne Musumba presented “How to Note: Capacity Building in Research” [1]. It described the detailed design of capacity building with research capacity building cycle. The weakness of this study is on capacity assessment to be effectively and assessment on target research duration. It did not considered on capacity assessment on target research area or field.

Laura F. Huenneke presented “Key Strategies for Building Research Capacity of University Faculty Members” [4]. In this study, it describes the key strategies such as: Promotion and Tenure Applications, Continued Employment at the University, Peer-Reviewed Publications, External Grant Proposals and Funding, Recruiting New Faculty from outside the Institution, Cultivating and Attracting Researchers from within Existing Faculty, Call for Proposals for Pilot Projects, Summer Research Conference, Investing in Research Infrastructure, Investing in Institutional Leadership and Planning for Leadership Succession and Transition. This study identified the research capacity building for university faculty member but it does not consider on capacity assessment on existing capacity of faculty members.

In “Research Capacity Building for Development: From Old to New Assumptions” [5], LÉA VELHO described to build research capacity by training Masters and Ph.Ds.: underlying assumptions. It

is effective only for Universities that are permitted to launch Masters and Ph.Ds.

4. RESEARCH DESIGN AND METHODOLOGY FOR RESEARCH CAPACITY BUILDING

The study describe by comparing the earlier RCB process with new proposed strategy in two articles.

4.1. Weakness of Earlier RCB

Earlier duration, strategy of RCB was not introduced finitely. According to the behavior of Research capacity building, RCB process in earlier duration are indicated as follows:

- I1. A rector to develop University Research
- I2. A research leader
- I3. Research Funding
- I4. Supporting human resource for collecting research data
- I5. Promoting Technical skills of research members
- I6. Motivating research members

According to the research behaviors, Research Performance indicators are as follows:

- P1. Research proposal
- P2. Research data
- P3. Digitizing research data
- P4. Preparing standard data
- P5. Analyzing Results
- P6. Willingness of research members

In the earlier duration, the research members were enhancing the existing basic technical skills to promote the performance of the research with the lack of capacity assessment to target research area.

Table 4.1. Percentage of each RCB process during research

Duration (MM/DD/YY)	I1	I2	I3	I4	I5	I6
6/10/17	100	100				
6/17/17						
10/27/18			100			
12/1/18				100		
2/1/19						
5/27/19						
6/30/19					50	100
7/30/19					50	
8/30/19						
9/30/19						
Total	100	100	100	100	100	100

Table 4.1 shows the percentage of each RCB process indicator and table 4.2 shows the percentage of each performance during research.

Table 4.2. Percentage of each performance during research

Duration (MM/DD/YY)	P1	P2	P3	P4	P5	P6
6/10/17						
6/17/17	100					
10/27/18						
12/1/18						
2/1/19		100				
5/27/19			20			
6/30/19			20	40	0	20
7/30/19			20	20	10	20
8/30/19			20	20		
9/30/19			20	20		
Total	100	100	100	100	10	40

Table 4.3. The relation between RCB Process Indicators and RPB Indicators

NO.	Fulfilling Capacity Building Process	Research Performance getting due to Capacity building process
1	I1	P1
2	I2	P2
3	I3, I4	P2, P3
4	I5	P4
5	I6	P4, P5, P6

Table 4.3 summarizes the relation between RCB process indicators and performance indicators.

As show in table 4.1 and 4.2, the rector announced to develop university research: I1 in 6 October 2017. It is the start and first of all capacity building process. As a consequence, the head of one department proposed the research proposal. This is the first research performance, P1, getting by fulfilling of first capacity building, I1. The percentage of these two indicators I1 and P1 are both 100% as shown in tables.

As the second RCB process, I2, the department head start to lead the research. The performance indicator, P2, due to the I2 is the research data need to collect during 17 October 2017. Both are also 100%.

The third RCB process is to fulfill research funding, I3, and to support human resource to collect research data, I4. These capacity building processes were finished during the year 2018. After 2018, the performance of these capacity building is getting the collected research data. Indicator P2 is completed. The next performance P3 is digitizing research data.

Fourthly, capacity building process, I5, promoting technical skill on research members is started. In this process, there was no capacity assessment on each research member upon target research area. Due to this weakness, although the percentage of performance P4 is 100%, the percentage of performance P5, analyzing the result, is just 10%. As a consequence, P6, the wiliness of research members is also about 40%.

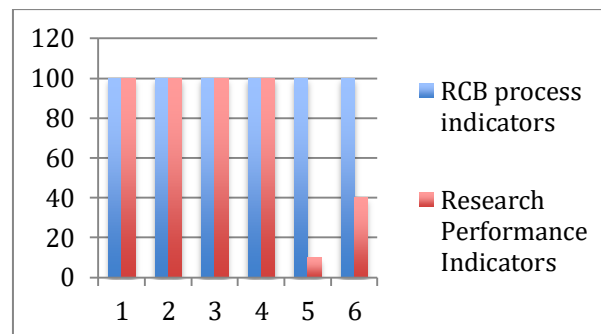


Figure 4.1. Comparisons of RCB process indicators and research performance indicators of earlier RCB

Figure 4.1 shows that the two performance indicators: P5, analyzing results and P6, willingness of research members did not improve. The RCB indicators and RP indicators were not balancing.

4.2. Research Design and strategy for RCB process

The research design and strategy emphasized on the first step of research capacity building process: capacity assessment according to the weakness of the old strategy. The proposed RCB cycle is shown in Figure 3.2.

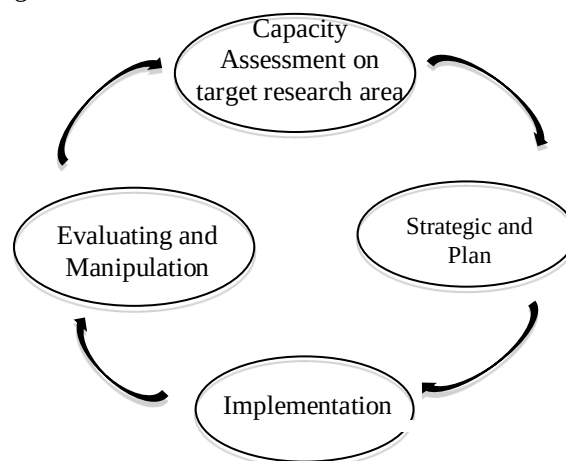


Figure 4.2. Proposed RCB cycle

The proposed RCB cycle is modified in capacity assessment. Initial work involves understanding the gaps and needs, and then setting priorities for research capacity building.

This first stage must start with an appraisal of

the current research and institutional framework; it is an essential part of the inception phase. Such an analysis should focus as much on identifying assets, or areas of relative strength, as weaknesses. In general, it is easier to build on the former than to transform the latter. This analysis should include an assessment of the position at the three levels: individual, organizational and institutional, and should, as far as possible, be conducted as a collaborative exercise with key partners. Capacity building begins with a shared and explicit awareness of individual, organizational and institutional strengths and weaknesses in all partner teams and organizations.

Individual: Training needs—skills and knowledge gaps between existing skills and target research area

Quality of research output

Organizational: Systems and resources on target research area

Institutional: Stakeholder analysis on target research area

In the second stage, according to the capacity assessment on target research area, planned the detailed activities to build the technical skills for individual members required to deliver the program outcomes. Thirdly, implementation starts for all three levels such as individual, organizational and institutional. Fourthly, The same balance to be struck between “process” and “performance” was highlighted in the planning and implementation stages, also needs to be present in Monitoring & Evaluation.

Actually, the fourth step is not a final step, after that, RCB cycle is started again.

5. MONITORING AND EVALUATION OF PROPOSED RESEARCH DESIGN AND STRATEGY

The study emphasized the importance of a process approach to building capacity. While it is need to recognize the centrality of learning, adaptation and power relations in CB processes, it is also vital to recognize the need for performance. Performance is an indicator of capacity and without performance capacity building is rather pointless. It is therefore necessary to seek a balance between the importance of process and the critical nature of results and outcomes in the approach.

The findings shows that during capacity assessment step, the existing capacity of research members is not match with the target research proposal and field. Promoting the technical skills of individual members on target area has a positive effect on the result performance.

RCB process in proposed strategy are indicated as follows:

- I1. A rector to develop University Research
- I2. A research leader
- I3. Research Funding
- I4. Supporting human resource for collecting research data
- I5. Promoting Technical skills of research members on target research area
- I6. Motivating research members

Table 5.1 shows the percentage of each RCB process upon capacity assessment on target research area.

Table 5.1. Percentage of each RCB process during research

Duration (MM/DD/YY)	I1	I2	I3	I4	I5	I6
6/10/17	100	100				
6/17/17						
10/27/18			100			
12/1/18				100		
2/1/19						
5/27/19						
6/30/19						
7/30/19						
8/30/19					100	100
9/30/19						
Total	100	100	100	100	100	100

The research performance indicators are promoted as shown in table 5.2.

Table 5.2. Percentage of each research performance (RP) during research

Duration (MM/DD/YY)	P1	P2	P3	P4	P5	P6
6/10/17						
6/17/17	100					
10/27/18						
12/1/18						
2/1/19		100				
5/27/19			20			
6/30/19			20	40		
7/30/19			20	20		
8/30/19			20	20		
9/30/19			20	20	60	100
10/10/19					40	
Total	100	100	100	100	100	100

The following Figure 5.1 shows that after fulfilling capacity assessment and technical training on target research area, all members are willingness to research and the analysis results of the research outcomes. The findings are also evidence that there is a balance between RCB process and research performance of proposed strategy in research capacity building.

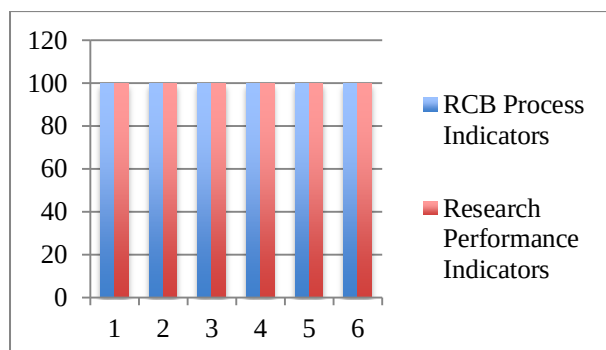


Figure 5.1. Comparison of RCB process indicators and RP indicators of proposed strategy

6. CONCLUSION

The study highlighted that if the results were not produced while RCB was an ongoing process, capacity assessment on this process could be the vital role in RCB. Because of the variance in existing capacity of research members, the initial work should involve capacity assessment on target research area to occur balancing between process and performance. The study examined that if the UR has a systematical strategy for RCB, the performance of the research will promote in time. As a Consequence, the proposed RCB cycle can motivate the UR members' willingness in research.

ACKNOWLEDGEMENTS

The author would like to express special thanks to her colleagues from ITSM department for their participating, analyzing and assessment on research's performance.

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Impact of Written Exam on Students' Project Performance

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ABSTRACT: *This paper studied the students' performance on project based learning for each academic year of University of Computer Studies (Taungoo). The study analyzed the assessment of subjects for first year, second year and third year computer science students based on written exam and no written exam from 2017-2018 academic year to 2018-2019 academic year. Then, percentage of student based on assessment of project using defined rubric was mathematically calculated and then displayed bar chart to compare two academic years. Result of research indicates that programming language subjects using project based learning not including written test has positive impact on students' project performance.*

Keywords: Project Based Learning, Written exam, Assessment, Performance

1. INTRODUCTION

The Computer Science (CS) academic community regards group-work project as an essential component of any degree; the discipline's professional societies world-wide emphasize project and group work as preparation for professional practice. Project work is recognized as having many educational and social benefits. The lessons learned in a particular institution are rarely generalized and the good practices are rarely disseminated[5].

All Computer Universities incorporate project work into their degree and many have evolved imaginative and effective ways of managing student project. Most Computer Science departments use project based learning for some of computer science subjects.

1.1. Research Motivation

The author has been teaching one subject of computer programming language in two academic years. This subject has assessment methods such as practical, class participation, assignment, project, lab exam and written exam. But the assessment of subject is different for two academic years. In current academic year, there is no written test. The assessment of project improves year after year depending on not having written exam. Therefore this research analyzes the improvement of project assessment for students for two academic years.

1.2. Research Objective

The main purpose of this study is to enhance the students' performance on project work in programming language subjects of University of Computer Studies (Taungoo). It analyzes three computer programming language subjects of ITSM department for two academic years. The subjects are C#, JavaScript and Hyper Text Markup Language (HTML).

Computer science subjects use assessment

methods and learning methods to provide knowledge, problem solving, communication, critical thinking, creativity and performance.

Firstly, finding of the current study shows the percentage of assessments for three subjects based on project based learning. Secondly, the subject of teaching was practical, whenever the assessment was based on written test in previous academic year. This subject more needs to test practical ability. So, written test does not include in current academic year.

Finally, the study indicates that assessment of project not including written test has a better performance on current academic year.

2. LITERATURE REVIEW

Computer based testing is a very convenient method. In addition to this, it increases the transparency of marking and students get the freedom to develop the application using their own concepts and methods. It develops their creativity as well as thinking power. Students more emphasize on project work and learn more. They do not need to remember unnecessary theories or memorize some points. As we know when we are in front of the computer we will remember what to do next automatically compared with when answering a paper using a computer[6].

Another reason is that reading coding is very hard because most of their hand writing is not so neat. Since there is no way to debug their coding there are so many errors and mistakes. They memorize program code for written exam and they emphasizes on written exam marks to get better grade. According to study, they also prefer practical based testing than written exam[1].

Project Based Learning has become a widely used method of teaching in technical subjects. Project-based learning is a model that organizes learning around projects. Project-based learning emphasizes learning activities that are long-term, interdisciplinary and student-centered.

Typical projects present a problem to solve (What is the best way to easily search a book in the

university library?). To solve this, students create computerized library management system for administrator and students. PBL intentionally develops students' problem solving and creative making of publicly presented product or performance

Project-based learning also gives students the opportunity to explore problems and challenges that have real-world applications, increasing the possibility of long-term retention of skills and concepts.

Variance is one of the most useful tools in probability theory and statistics. In science, it describes how far each number in the data set is from the mean. In practice, it often shows how much something changes.

3. RELATED WORK

In "A Comparison Between Evaluation of Computer Based Testing and Paper Based Testing for Subjects In Computer Programming", by Samantha Mathara Arachchi, the students' questionnaire 85% said that computer based testing is much better and they have proved it is more successful than the paper based testing due to the following advantages; No need to by-heart concepts and unnecessary theory, Reduce stress, Build more confidence about the answers, The answer is visible, Working in front of computer is more comfortable, Interface provides the extra support, Ability to debug time to time to make it correct, More freedom to come up with alternative methods and Easy to score maximum marks[6].

"Students' Perceptions of Paper-Based vs. Computer-Based Testing in an Introductory Programming Course" by Jo~ao Paulo Barros examines the preferences of students regarding computer-based versus paper-based assessment, in an introductory computer programming course. Two groups of students were surveyed about their preference between paper-based and computer-based tests and respective rationale. All students had already been assessed: one group using two paper-based tests and the other group using two computer-based tests. Both groups expressed an overwhelming preference for computer-based tests independently of their previous programming experience[1].

In "Teaching Methods and Its Efficacy An Evaluation by the Students", by Shreemanta Kumar Dash, the study was undertaken to elicit the perception of the MBBS students of KIMS, Bhubaneswar regarding the prevailing system of teaching methods, to evaluate the extent to which they are benefited by these and to obtain the views and suggestions of students about various aspects of the current medical education. 337 students were selected from 4 semesters for this study. A set of questionnaires were put and directed to choose the appropriate option as per the likert scale. The response were collected and analyzed. 51.83% of the total respondents agreed that topics during normal lectures are interesting, 77.02% stated normal lecture as the most effective method of teaching followed by

group discussion 68.02%. 67.07% are of the view that the faculty members are having good communication skill and 62.96% of them found that the faculty members are approachable, 82.83% of the participants agreed that black board teaching as the better mode of teaching, so also 55.87% agreed that audio visual aids as a useful mode of teaching, 56.10% of the participants agreed that the number of practical (clinical) classes should be more than the theory classes[7].

4. RESEARCH DESIGN AND METHODOLOGY

In University of Computer Studies (Taungoo), there are two major: Computer Science and Computer Technology. Faculty of Computer Science (FCS), Faculty of Information Science (FIS), and Information Technology Supporting and Maintenance (ITSM) departments are main major departments for Computer Science.

The computer programming language subjects: CSharp for third year, Java Script for second year and Hyper Text Markup Language for first year are subjects of ITSM departments. These subjects were coded as C#, JS and H.

The assessments of these subjects were collected for two academic years of University of Computer Studies (Taungoo). The study collects assessments of these subjects on almost 300 students of first year, second year and third year from 2017-2018 academic year to 2018-2019 academic year. In 2018-2019 academic year that did not have written exam although written exam have in 2018-2019 academic year.

There are five assessments for each subject as following.

- A1: Assessment of practical
- A2: Assessment of assignment
- A3: Assessment of lab test
- A4: Assessment of project
- A5: Assessment of class participation

Table 4.1. Percentage of assessment on each subject

Academic Year	2018	2019	2018	2019	2018	2019
Subject	H	H	JS	JS	C#	C#
A1	6	7	7	8	7	9
A2	6	6	7	7	6	8
A3	8	8	6	7	7	8
A4	6	8	6	8	7	9
A5	5	7	6	7	6	7
Total	31	36	32	37	33	41
Total %	62%	72%	64%	74%	66%	82%

The assessment test scores are given using the defined rubric for these subjects. These scores were mathematically categorized, coded and quantized by

four variables: “Bad”, “Natural”, “Good” and “Better” and ranging from “3-4”, “5-6”, “7-8” and “9-10” respectively. The scores of assessment are displayed for three subjects on having written exam and not. Percentage of assessment are displayed and compared for two academic years as shown in table 4.1.

The seven points are defined to give scores for project assessment of three subjects as following.

- P1. Description
- P2. User Interface Design
- P3. Implementation
- P4. Testing
- P5. Presentation
- P6. Participation
- P7. Question and Answer

The study collects data from appropriate authority. The study shows that the percentage of third year computer science students getting scores by Bad, Natural, Good and Better for C# subject and shows comparison of two academic year having written exam and not.

5. RESEARCH FINDINGS

Finding the variance on assessments of three subjects in two academic years is shown in table 5.1. Percentage of assessment on each subject in 2017-2018 has variance with 2018-2019 academic year.

According to following result, the percentage of assessments of project (A4) is the maximum value (0.72) in all assessments. The variance for assessment of project (A4) is the main point the makes the changes depending on not having and having written exam.

Table 5.1. Variance percentage of subject between two different years

Assessment	HTML Variance	JS Variance	C# Variance	Total Variance
A1	0.14	0.13	0.22	0.49
A2	0.00	0.00	0.25	0.25
A3	0.00	0.14	0.13	0.27
A4	0.25	0.25	0.22	0.72
A5	0.29	0.14	0.14	0.57
Total %	0.14	0.14	0.20	

Table 5.3 highlighted that there are no “Bad” condition of students in both years. In 2018-2019 academic year (no written exam), the number of “Natural” condition of students are down to (-24.3 %) rather than that of previous year (with written exam). This is because of the improvement of number of students in both “Good” (+8.6 %) and “Better” (+22%) condition. The findings show that students were emphasizing in skills of coding and critical thinking of

the subjects rather than in getting written exam marks by studying by-heart.

Table 5.2. Mean value for number of student involvement in C# subject between two different years

Proj :Ass :	2017-2018 Academic Year				2018-2019 Academic Year			
	Bad	Natural	Good	Better	Bad	Natural	Good	Better
P1	0	70	30	0	0	60	40	0
P2	0	60	40	0	0	30	40	30
P3	0	50	50	0	0	40	50	10
P4	0	80	20	0	0	40	30	30
P5	0	40	60	0	0	30	60	10
P6	0	80	20	0	0	30	40	30
P7	0	60	40	0	0	40	60	0
Mean	0	62.9	37.1	0	0	38.6	45.7	22

Table 5.3. Mean comparison of number of student involvement in C# subject between two different years

	Bad	Natural	Good	Better
Mean compare	0	-24.3	8.6	22

Figure 5.1 shows the Comparison of assessment of project for HTML subject in two academic years. In this figure, there is “Bad” condition of students in both years. These students are first year students who have no experience with programming language subjects and project. But the students improve in “Good” condition in 2018-2019 academic year (no written exam). Some students have in “Better” condition in 2018-2019 academic year.

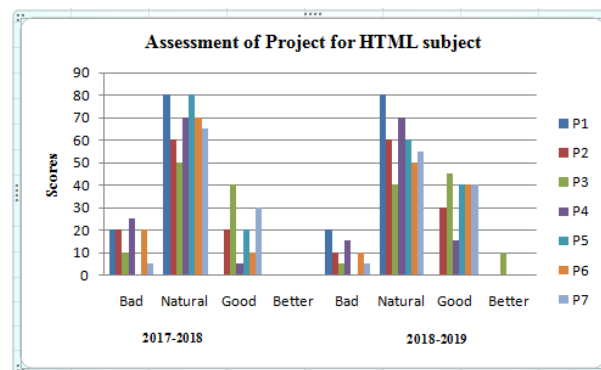


Figure 5.1. Comparison of assessment of project for HTML subject in two academic years

Figure 5.2 shows that Comparison of assessment of project for Java Script subject in two academic years. In this figure, “Bad” condition of students reduces in both years because they have

experience of project in first year. There is “Better” condition because of getting more practical skill.

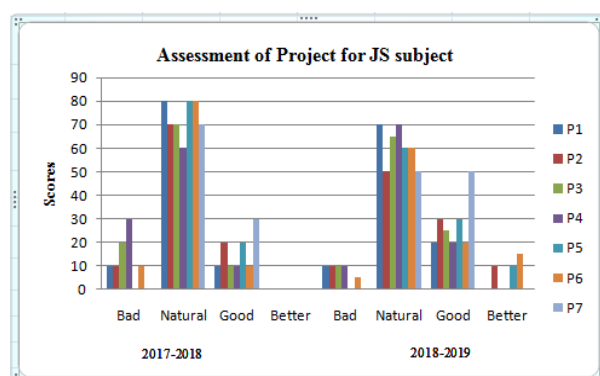


Figure 5.2. Comparison of assessment of project for Java Script subject in two academic years

Figure 5.3 shows that Comparison of assessment of project for CSharp subject in two academic years. In this figure, there is no “Bad” condition of students in both years and improve “Natural” condition to “Good” condition in 2018-2019 academic year (no written exam). There is “Better” condition in 2018-2019 academic year because they emphasize on skills of coding and critical thinking of the subjects.

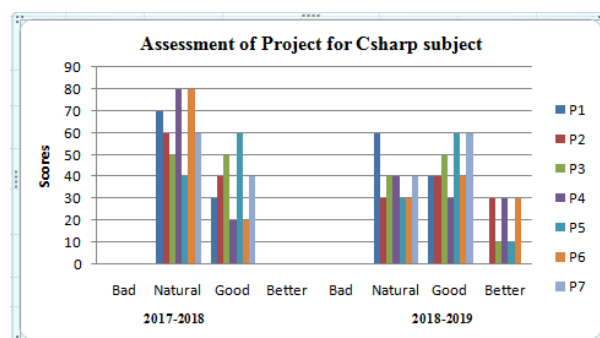


Figure 5.3. Comparison of assessment of project for CSharp subject in two academic years

6. CONCLUSION

The research data were evaluated and displayed by using two facts: Assessments of computer programming language subjects using Project Based Learning and Assessments of Projects for three subject depend on having written exam and not having written exam. The study shows that students were emphasizing in skills of coding and critical thinking of the subjects rather than in getting written exam marks. The study established that Assessments of Projects improve year after year. It is also evidence that some computer programming subjects need practice and creativity rather than to emphasize on written exam.

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A Study of Quality Assurance in Universities

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ABSTRACT: *The terms assessment, evaluation and quality assurance have various interpretations in universities. Quality is a multidimensional concept that touches not only upon quality assurance procedures, but also accessibility, employability, academic freedom, public responsibility for higher education in universities. It serves multiple purposes, such as enhancing learning and teaching, building trust among stakeholders throughout the higher education system and increasing regional and international harmonization and comparability.*

Keywords: *Assessment, Quality Assurance, Accessibility, Employability, Higher Education System*

1. INTRODUCTION

The terms assessment, evaluation and quality assurance have various interpretations in higher education. The first two, assessment and evaluation, share considerable conceptual ground and interconnected histories. Quality assurance, on the other hand, is a more recent development. The issue of academic achievement standards in particular has significant implications for quality assurance. Quality assurance (QA) is one aspect of quality management. Below are ISO 9000 definitions from ISO 9000:2015: Quality management systems - Fundamentals and Vocabulary. Quality assurance can be defined as "part of quality management focused on providing confidence that quality requirements will be fulfilled." The confidence provided by quality assurance is two folds—internally to management and externally to customers, government agencies, regulators, certifiers, and third parties. An alternate definition is "all the planned and systematic activities implemented within the quality system that can be demonstrated to provide confidence that a product or service will fulfill requirements for quality. "The Quality Office manages the Quality Assurance System in various fields of teaching departments in universities. Research and School Governance and improves it through internal and external reviews on a regular basis. Its goal is to provide a continuous enhancement of quality aiming at a culture of excellence within the whole institution for the accomplishment of its missions.

2. QUALITY MANAGEMENT SYSTEMS AND STANDARD OPERATING PROCEDURES

A quality system is defined as the organizational structure, responsibilities, processes, procedures and resources for implementing quality management. Quality Management System (QMS) determines and implements the universities quality

policy and quality objectives. Top management commitment and active involvement in the establishment, management and monitoring of quality systems is critical and is achieved by;

- Defining and documenting a quality policy and quality objectives and ensuring that both the policy and objectives are understood and implemented by all employees at all levels;
- Ensuring that appropriate processes are implemented to fully satisfy customer needs and expectations and university objectives;
- Defining and documenting the responsibility, authority and interrelation of key personnel managing the quality systems;
- Providing adequate resources for implementing and maintaining the quality systems;
- Conducting scheduled management reviews of the quality systems to assess their continued suitability, adequacy, effectiveness and efficiency; and
- Deciding on actions for continual quality improvement.

2.1 Internal Quality Assurance (IQA) Unit of Universities

Quality Assurance refers to the process of maintaining standards reliably and consistently by applying criteria of success in a course, programme or institution. Quality Assurance can be defined in terms of its four components:

- (a). everyone in the enterprise has a responsibility for enhancing the quality of the product or service;
- (b). everyone in the enterprise has a responsibility for maintaining the quality of the product or service;
- (c). everyone in the enterprise understands, uses and feels ownership of the systems which are in place for maintaining and enhancing quality; and
- (d). management regularly checks the validity of the system for checking quality.

Every educational institution is a dynamic system and must be seen in its own uniqueness and totality for quality management. In order to look into the different

aspects of quality of higher education institutions, it is essential to understand the systems approach to education. The notion of a system is a theoretical abstraction, and can be defined as an organized unitary whole composed of a set of interdependent, interrelated and interacting synergetic elements or sub-systems and delineated by indentifiable boundaries from its environmental supra-system.

Areas of concern for IQA

- Teaching
- Learning
- Research
- Services
- Management

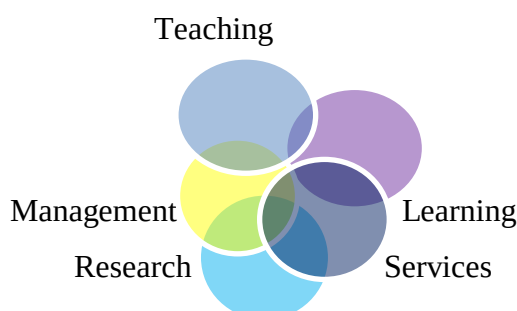


Figure 1. Areas of Internal Quality Assurance

2.2 Standard Operating Procedures

Standardization is defined as an activity that gives rise to solutions for repetitive application to problems in various disciplines including science and it is aimed at achieving the optimum degree of order in a given context. Generally, the activity consists of the process of establishing (determining, formulating, and issuing) and implementing standards. Therefore, standards are the ultimate result of a standardization activity and within the context of quality systems consist of quality documents or documents related to the quality systems.

The quality documents consist of Company policies, quality management plan, SOPs, working instructions, conventions, guidelines, forms, templates, logs, tags and labels. They are established by consensus and approved by a nominated body and they provide for common and repeated use, rules, guidelines or characteristics for activities or their results with a view to promote transparency, consistency, reproducibility, interchangeability and to facilitate communication. The hierarchy and types of quality documents relevant to quality systems will depend upon Company business objectives and business model. SOPs are Level 2 quality documents and, along with other relevant quality documents, ensure the effectiveness and efficiency of quality systems.

3. MANAGEMENT OF QUALITY ASSURANCE

Quality assurance procedures form part of a coherent system, which is central to the promotion of a quality assurance culture within the provider. The basic activities captured in the policies and procedures of the quality management system include, but are not limited to, the following:

- A description of processes.
- Identification of roles and positions responsible for the implementation of quality assurance policies and procedures, which are clearly described and designated.
- Self-monitoring of processes. Indicators are developed and data is collected which can be used to measure the effectiveness of policies and procedures.
- Systematic analysis of QA findings following self-evaluation involving stakeholders. Layers of moderating and supervisory processes, as appropriate, are in place to enable strategic and co-ordinated analysis of information at appropriate levels.
- Clear decision-making and follow-up. Systematic formal decision-making procedures are built into a provider's various evaluative mechanisms, including the procedures for oversight of such mechanisms by the governance system or equivalent. Management acts on the self-evaluation findings.
- A resource base which is sufficient to ensure sustainability. The corporate infrastructure, including financial solvency of the business, supports the sustainability of the teaching and learning environment and underpinning quality assurance system. The provider is adequately resourced to undertake and complete the education and training or research programmes proposed. Continuity planning procedures are developed to understand and anticipate events that could threaten the provider and to ensure that the provider remains viable.

3.1 Examples of Areas in a Provider's QA System

Some examples of areas to be included in a provider's QA system include:

- The organization's Mission to clarify its role as a provider of education and training programmes.
- Policies showing full understanding of the legislative obligations of the provider. The policies will inform:
 - (i). Management and staff as to the general approaches to follow in their work
 - (ii). Learners and other stakeholders as to what they can expect of the provider
- The procedures designed to implement the policies.

An internal self-monitoring system which will regularly check the effectiveness of the procedures and act as an early warning system in areas of provision requiring improvement. This includes the use of data collected and compared against indicators, for example, completion and progression rates for learners.

A self-evaluation system through which the provider will review and report on the quality of its own programmes and related services. Each such evaluation will include the views of learners and independent evaluators who can make comparisons with other similar programmes offered elsewhere.

A system for responding to the findings of self-monitoring and review which will address areas for improvement and build on areas of strength. The provider uses indicators/benchmarks against which programme data can be compared.

4. SELF-EVALUATION, MONITORING REVIEW

Review and self-evaluation of quality, including review of programmes of education and training, research and related services, is a fundamental part of the provider quality assurance system. The purpose of such self-evaluation is to review, evaluate and report on the education, training, research and related services provided by the provider and the quality assurance system and procedures which underpin these. In doing so, existing effective practices are identified and maintained, while areas needing improvement are addressed.

4.1 Provider-Owned Internal Review, Self-Evaluation and Monitoring

The outcomes of provider-owned and external reviews and follow-up actions taken are considered by the provider when preparing for upcoming external reviews. The findings of self-evaluations are analyzed and are available for, and connect to and support, the external review cycle. The distinction between ongoing, internal self-monitoring and formal self-evaluation is typically in frequency and scale. Self-evaluation has a broad, systemic focus and is carried out at specified intervals. Internal self-monitoring is ongoing and typically focuses on specific indicators.

4.2 Internal Self-Monitoring

Self-monitoring procedures include:

- (i) A system of appropriate quality measures: Appropriate quality measures are identified which can be checked in monitoring, such as, learner satisfaction ratings, completion rates, relevance of outcomes to the market place, error levels. The quality system monitors key performance indicators and progress against objectives.
- (ii). Gathering evidence of achievement of objectives: When objectives are set, consideration is given to the kind of evidence or information required to determine whether or not the objective has been met, and how and where this information will be obtained and by whom. Outcomes-related objectives (i.e. product) are especially important, but objectives concerning inputs, processes

and environments are also part of the quality assurance system.

(iii). Consideration for prioritising objectives: important objectives are prioritised even where they are difficult to achieve or to quality assure. Quality assurance is not limited to addressing easily assessed objectives nor committed to addressing inconsequential ones.

(iv). Acting on findings: All provider-owned quality assurance evaluation outcomes are used to produce a quality improvement plan which sets out a schedule of actions to be undertaken following internal evaluation. It identifies the persons responsible for actions and follow-up. Quality assurance does not produce perverse incentives.

5. CONCLUSION

In conclusion, this paper has presented an overview and a brief literature review of the main aspects related to QA in universities. Indeed, Quality Assurance (QA) is a term generally used to describe the processes that seek to ensure that the learning environment such as including teaching and research reaches an acceptable threshold of quality. QA is also used to describe the enhancement of education and training provision and the standards attained by learners. There is no agreement as to a QA definition or a QA model. In this study, we also discussed various issues related to understanding of quality assurance in universities. Higher education is at the cross roads. At one end there is high demand for access to higher education, and at the other the quality is questioned. In order to survive in the competitive world of globalization, all higher education institutions should pay special attention to quality in higher education. A challenge revolves around the concerns of faculty members and stakeholders, such as students, about the QA process. Students are the center of higher education, and invest time and money in the system, consisting them could improve QA processes.

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